COURSE TITLE: *Python Programming*			
Course Code:	**24PCA203**	**Course Type:**	**CC**
Teaching Hours/ Week (L:T2:P):	**3:0:2**	**Credits:**	**4**
Total Teaching Hours:	**45+0+30**	**CIE + SEE Marks:**	**50+50=100**

***Preamble** (brief description).*

Understanding why Python is a useful scripting language for developers begins with Python installation and setup, which is straightforward and accessible. Python's simplicity and readability make it an excellent choice for scripting. Developers can efficiently use different conditional statements such as if, if-else, elif, nested loops, and while and for loops to control the flow of their programs.

Course Outcomes

At the end of the course students will be able to…

CO1	Recognize and use different data types, convert data types and will be able to create and manipulate strings in Python.
CO2	Recall the basic knowledge of modules, functions, and packages and be able to create their own modules and packages in Python.
CO3	Analyse the different options in Data Management in Python Programming Language.
CO4	Implement exception handling in Python, including using try-except blocks, raising exceptions, and handling file-related exceptions.
CO5	Apply their skills in a real-world scenario by designing and implementing a program that handles file operations and exceptions in Python.

PO – CO Mapping

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**
CO1	3	3	2	3	3	-	-	2
CO2	2	3	3	2	2	-	-	3
CO3	1	2	3	2	3	-	-	3
CO4	3	2	2	2	3	-	-	2
CO5	3	3	3	3	3	-	-	3

Note: ‘-‘=Not Relevance; ‘1’=Low Relevance; ‘2’ = Medium Relevance; ‘3’=High Relevance

Course content

Theory

Unit 1 **09 Hours**

An Introduction to Python:

History and development of Python, Why Python? Grasping Python’s core philosophy – Installation - Python Interpreter – Interpreter and its environment. Working at the command line or in the IDE, Installing Anaconda on Windows, Linux and MAC, Syntax and Indentation.

Online IDE: Introduction to Google colab note book, Setting up a new Colab notebook.

Basic Python: Print statement, Comments, Variables, Assignments, Numbers and Operators.

Data types: Numeric, Dictionary, Boolean, Set, Sequence and type.

Type Conversion: Implicit and explicit conversion (casting). Creating and using String.

Hands-on / Case Study: **(Requirements: Any Python *IDE*)**
1. **Installing Python**: Install Python on your computer and test that it is working correctly by opening the Python interpreter and executing some basic commands.
a) Using an IDE: Install and use an integrated development environment (IDE) such as PyCharm or Visual Studio Code. Create a new Python project, write some code, and execute it.
b) Basic Python Operations: Write a Python program to print "Hello World!" on the screen. Write a program to perform basic arithmetic operations such as addition, subtraction, multiplication, and division. *(Hands-on)*
2. **Data Types:** Crete a Student Grade Tracker system using python that uses various data types such as strings, integers, floats, booleans, lists, and dictionaries to implement these functionalities. *(Hands-on)*

Unit 2 **09 Hours**

Flow Control in Python:

Python Statement: If Statement, If else statement, elif statement, short hand if, else statement, While Statement, Definite Loops vs Indefinite Loops, Nested Loops, Abnormal Loop Termination, While/else and for/else, for statement, Range statement.

Conditional expression: AND, OR, NOT.

Python Conditions: Nested Conditions using all conditional statements, Break and continue using all conditional statements statement, Python pass.

Data structures: List, tuple, string, set and dictionary.

Hands-on / Case Study: **(Requirements: Any Python IDE)**

1. Write a program that prompts the user to enter a number, and then prints "Positive" if the number is greater than zero, "Negative" if the number is less than zero, and "Zero" if the number is equal to zero. *(Hands-on)*
2. Write a program that prompts the user to enter a password. If the password is "password123", the program should print "Access granted". Otherwise, the program should print "Access denied". *(Hands-on)*
3. Write a program that prompts the user to enter a number, and then prints the factorial of that number using a while loop. *(Hands-on)*
4. a) Write a program that prints the first 10 numbers of the Fibonacci sequence using a for loop.
   b) Write a program that prompts the user to enter a string, and then prints each character of the string on a separate line using a for loop.
   c) Write a program that prints the first 10 even numbers using a while loop. *(Hands-on)*
5. Write a program that prompts the user to enter two numbers, and then prints the larger number using a conditional expression. *(Hands-on)*
6. Write a program that prompts the user to enter a number, and then prints "Positive" if the number is greater than zero, "Negative" if the number is less than zero, and "Zero" if the number is equal to zero, using a conditional expression. *(Hands-on)*

Unit 3 **09 Hours**

Functions module and package:

Introduction to module: Creating a module, Importing a module, Advantage of modules.

Introduction to functions: Calling a function, Types of function, Arguments and Parameters, *args and **kwargs, Lambda function and map, Array in Python.

Scope of Function: Global and local variable or Scope.

Insights of Function: Return function, Iterator function, Generator function, Difference between Iterator and generator, First class function, Python recursion(memorization).

Package: Introduction to package creating package, Date, Time, Math, Json, Regex, Pip.

Hands-on / Case Study: **(Requirements: Any Python IDE)**

1. Python function to check if a given number is prime or not. *(Hands-on)*
2. Python program to know how to define a function inside another function. *(Hands-on)*
3. Python program to understand the positional arguments of a function. (Hands on)
4. A Python program to access global variable inside a function and modify it. (Hands on)

Unit 4 **09 Hours**

Python - Object Oriented Programming

Introduction to Object oriented programming: Concept of OOP, Advantage of OPP, Class and Object.

Python Classes and Objects: Creating classes and objects, Class constructors and destructors, Instance variables and methods, Class variables and methods.

Inheritance: Creating a subclass, Overriding methods, Accessing parent class methods and attributes, Polymorphism.

Encapsulation: Access modifiers, Properties and getters/setters, Private and protected members.

Hands-on / Case Study: **(Requirements: Any Python IDE)**
1. Creating and using classes and objects:
    a) Create a class for a car with instance variables for make, model, and year.
    b) Create an object for a car and set its make, model, and year.
    c) Print the car's details using its instance variables. *(Hands-on)*
2. Write a program to access the base class constructor from sub class. *(Hands-on)*
3. Write a program to override super class constructor and method in sub class. *(Hands-on)*

Unit 5 **09 Hours**

File Handling

Introduction to File Handling: Types of files, Opening and closing files, Reading from a file, Writing to a file, File modes.

Reading and Writing Text Files: Reading a file line by line, Writing to a file line by line, Appending to a file, Using 'with' statement, simple input output function, output formatting.

Reading and Writing Binary Files: Opening binary files, Reading binary files, Writing to binary files, Pickling and unpickling objects.

Hands-on / Case Study: **(Requirements: Any Python IDE)**
1. File Handling Objective Tasks:
    a) Create a new text file and write some text to it.
    b) Open the file in read mode and display its contents.
    c) Open the file in append mode and add some more text to it.
    d) Open the file in write mode and overwrite its contents with new text. *(Hands-on)*

2. Reading and Writing Text Files Objective:
    a) Read a text file line by line and display each line.

b) Write text to a new text file line by line.
c) Append text to an existing text file.
d) Use the 'with' statement to open a file and read its contents.
e) Use the 'with' statement to open a file and write text to it. *(Hands-on)*c) Open the file in append mode and add some more text to it